is not compiled into native machine language. Instead, Java makes bytecode. Bytecode is the result of a "compile", but the compile results in an intermediate form that stops short of native machine-specific code. Instead of making a machine language native code for each particular OS, Java makes a single, universal bytecode module that feeds into any Java Virtual Machine (JVM). Each OS has its own different implementation of the Java Virtual Machine. The JVM sets up its own world within your RAM. It creates an internal software-only sub-computer within the OS. The bytecode talks to the JVM, and the JVM talks to the Operating System. So bascally "Write Once, Run Anywhere". The Virtual Machine interprets the bytecode one instruction at a time, and translates it into native machine code. You compile your program once into bytecode, but it is interpreted anew every time it runs.

In chapter 1, we cover the basic principles behind Java applications. Here we look at the way code is written to build applications.

In chapter 2, we discuss the various operators used in Java. Chapters 3, 4 and 5 discuss terms such as Methods, Objects and classes, and object oriented programming.

Data structure concepts such as Arrays and strings are discussed in chapter 6. Similarly, Files and streams are covered in chapter 7. Multithreading, a concept that enables multiple simultaneous lines of execution, is covered in chapter 8.

Chapter 9 and 10 is about graphics and GUI.